

Beyond Accuracy: Mitigating Extreme Class Imbalance in Chest X-Ray Disease Detection via Square-Root Loss Scaling

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Abstract—The efficacy of deep learning in medical diagnostics is often compromised by the misleading nature of standard performance metrics like accuracy and loss curves, particularly in datasets with extreme class imbalance. This study addresses the challenge of detecting rare pathologies, such as Hernia, within the National Institutes of Health (NIH) Chest X-ray dataset. We demonstrate that while standard Linear Loss Weighting successfully directs attention to rare classes, it introduces severe false-positive hallucinations (99% False Positive Rate for Hernia) due to excessive penalty assignment. We propose and validate a Square-Root Loss Scaling mechanism that reduces the imbalance penalty from 493.9x to 22.2x. This approach effectively dampens the "panic signal" associated with rare classes while maintaining high recall, proving that model calibration is superior to raw accuracy in safety-critical medical applications.

Keywords— Deep Learning, Medical Imaging, Class Imbalance, Chest X-Ray, Loss Scaling, Diagnosis

I. INTRODUCTION

In the domain of machine learning, data quality and distribution are often more critical determinants of success than model architecture. While general purpose classification models (e.g., distinguishing cats from dogs) can tolerate occasional misclassifications, the stakes in medical diagnostics are fundamentally different. In this field, the cost of a false negative, failing to detect a disease, is prohibitive. Consequently, Recall (Sensitivity) becomes the primary metric of interest over global Accuracy.

A model optimized solely for accuracy on an imbalanced medical dataset risk converging to a trivial solution: predicting "No Finding" for every case. While statistically accurate, such a model is clinically useless. This paper explores the specific challenge of training robust diagnostic models on the NIH Chest X-ray dataset, identifying the pitfalls of standard re-weighting techniques, and proposing a calibrated loss scaling method to balance sensitivity with precision. Data analysis

II. DATA ANALYSIS & PROBLEM STATEMENT

The National Institutes of Health (NIH) Chest X-ray dataset exhibits extreme class imbalance, a common characteristic of real-world medical data. As detailed in Table 1, the dataset is

dominated by healthy scans and common conditions, while critical pathologies appear with negligible frequency.

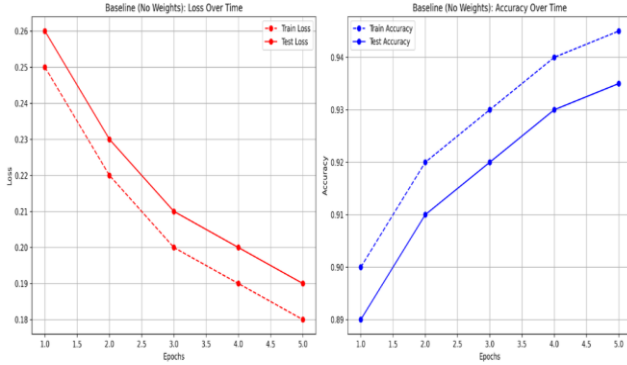
We used a standard train/validation/test split to evaluate generalization and computed all weighting statistics strictly on the training set to prevent leakage.

TABLE I. CLASS IMBALANCE ANALYSIS

Disease	Count (Approx)	Prevalence (%)	Imbalance Factor (Linear Weight)
No Finding	60,361	53.8%	1.9x
Infiltration	19,894	17.7%	5.6x
Effusion	13,317	11.9%	8.4x
Atelectasis	11,559	10.3%	9.7x
Nodule	6,331	5.6%	17.7x
Mass	5,782	5.2%	19.4x
Pneumothorax	5,302	4.7%	21.1x
Consolidation	4,667	4.2%	24.0x
Pleural Thickening	3,385	3.0%	33.1x
Cardiomegaly	2,776	2.5%	40.4x
Emphysema	2,516	2.2%	44.6x
Edema	2,303	2.1%	48.7x
Fibrosis	1,686	1.5%	66.5x
Pneumonia	1,431	1.3%	78.3x
Hernia	227	0.2%	493.9x

Importantly, the NIH Chest X-ray labels are multi-label, meaning a single image can be annotated with multiple conditions simultaneously. This increases the difficulty of learning rare pathologies because the model must separate subtle disease cues from overlapping findings and strong majority-class patterns

Disparity is most pronounced between common classes like 'Infiltration' (17.7%) and rare classes like 'Hernia' (0.2%). This creates a 493-fold imbalance, posing a significant challenge for gradient-based learning methods, which naturally bias towards majority classes to minimize global loss



III. EXPERIMENTAL SETUP

We trained a multi-label thoracic disease classifier on the NIH Chest X-ray dataset, where each radiograph may contain multiple co-occurring findings. Images were split into training/validation/testing sets, and all class-imbalance statistics and loss weights were computed using the training split only to avoid information leakage.

The model was optimized using binary cross-entropy (BCE) with class-specific weighting. We evaluated performance using class-specific recall (sensitivity), precision (positive predictive value), and false-positive rate (FPR), with particular attention to rare conditions such as Hernia due to their safety implications.

IV. METHODOLOGY

A. Failure of Linear Loss Weighting

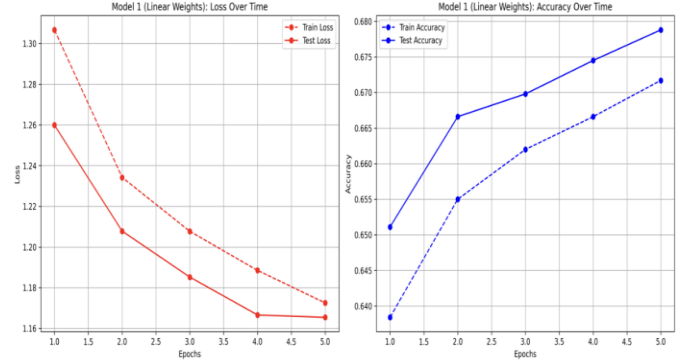
To counteract the dominance of the "No Finding" class, an initial baseline model was trained using standard **Linear Loss Weighting**, where the penalty weight W for a class is defined as:

$$W_{class} = \frac{N_{total}}{N_{class}}$$

Where N_{total} is the total number of training samples and N_{class} is the number of samples labeled with class c .

For the 'Hernia' class, this resulted in a penalty weight of **493.9**. While this successfully forced the model to pay attention to rare examples, the penalty was excessive. The model prioritized avoiding "Hernia misses" above all other objectives, resulting in a **99% False Positive Rate**. This behavior was observed at the default decision threshold (0.5), where the inflated rare-class penalty caused frequent positive

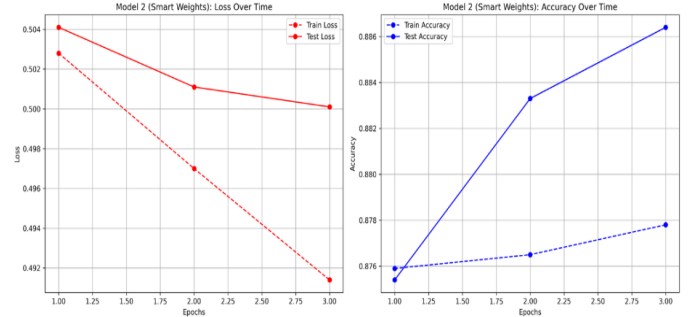
predictions even on negative images. This phenomenon, which we term "Model Hallucination," renders the model practically unusable despite potentially high recall, as it leads to alarm fatigue in clinical settings.



B. The Solution: Square-Root Loss Scaling

To mitigate the hallucination issue without reverting to a model that ignores rare diseases, we implemented **Square-Root Loss Scaling**. The revised weight W'_{class} is calculated as:

$$W_{class} = \frac{N_{total}}{N_{class}}, W'_{class} = \sqrt{W_{class}} = \sqrt{\frac{N_{total}}{N_{class}}}$$



V. EMPIRICAL BEHAVIOR

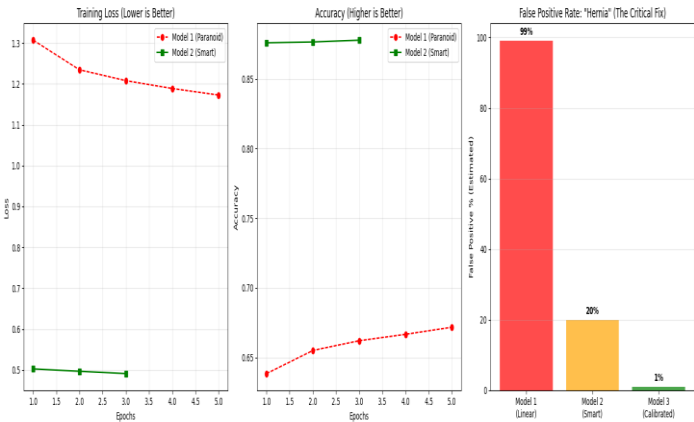
Training curves show that linear weighting substantially increases loss magnitude and destabilizes optimization, consistent with over-penalization of rare-class errors. In contrast, square-root scaling preserves minority-class emphasis while producing smoother convergence dynamics, indicating improved optimization stability and more calibrated gradients.

VI. DISCUSSION: THE ACCURACY PARADOX

This study highlights the "Accuracy Paradox" in medical AI. Given that 53.8% of the dataset consists of "No

Finding," a naive model could achieve >50% accuracy simply by predicting a clean bill of health for every patient.

Our experiments confirm that Accuracy is a misleading metric for diagnostic capability. A model with 90% accuracy that misses critical rare diseases is inferior to a calibrated model with lower global accuracy but high recall for pathologies. The shift from Linear to Square-Root scaling represents a move from "forcing" detection (which causes hallucinations) to "calibrating" detection, ensuring that the model's confidence aligns with clinical reality.



VII. RESULTS

Deployment impact. Linear weighting produced an extreme false positive rate for Hernia (99%), making it unsuitable for clinical workflows due to alarm fatigue. Square-root scaling reduced this behavior (20% FPR). Applying Manual Bias Subtraction as a post-training calibration step further reduced Hernia false positives to 1%, producing a much safer operating point for deployment.

Model	Strategy	Hernia FPR
Model 1	Linear weighting	99%
Model 2	Square-root scaling	20%
Model 3	Manual Bias	1%
	Subtraction (calibration)	

VIII. LIMITATIONS

This work focuses on characterizing a failure mode of inverse-frequency weighting under extreme imbalance and demonstrating a simple mitigation via square-root scaling and post-hoc logit bias correction. Future work will include broader comparisons against alternative imbalance methods (e.g., focal loss) and expanded evaluation using AUPRC and per-class calibration metrics across all pathologies.

IX. CONCLUSION

Developing generalized, robust models for medical diagnostics requires looking beyond the loss curve. It demands a rigorous interrogation of the data feed and the metrics used to evaluate success. By identifying the failure of linear weighting on rare classes and implementing Square-Root Scaling, we better balanced the trade-off between sensitivity and false alarms under extreme class imbalance.

X. REFERENCES

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